Introduction à la Programmation et l'Algorithmique

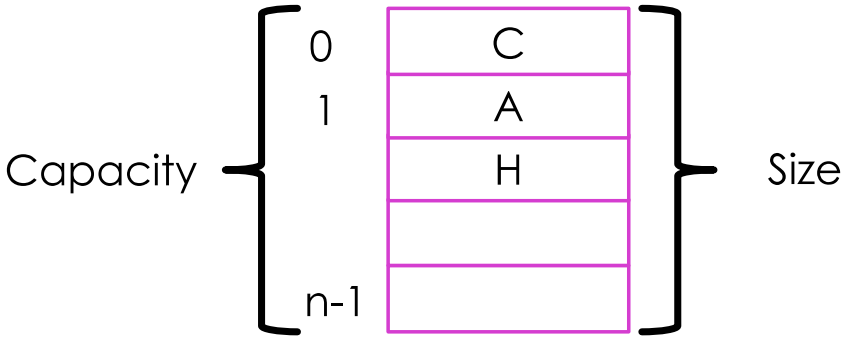
CHRISTELLE CAILLOUET – CHRISTELLE.CAILLOUET@UNIV-COTEDAZUR.FR

DINO LOPEZ PACHECO – DINO.LOPEZ@UNIV-COTEDAZUR.FR

Data Structures

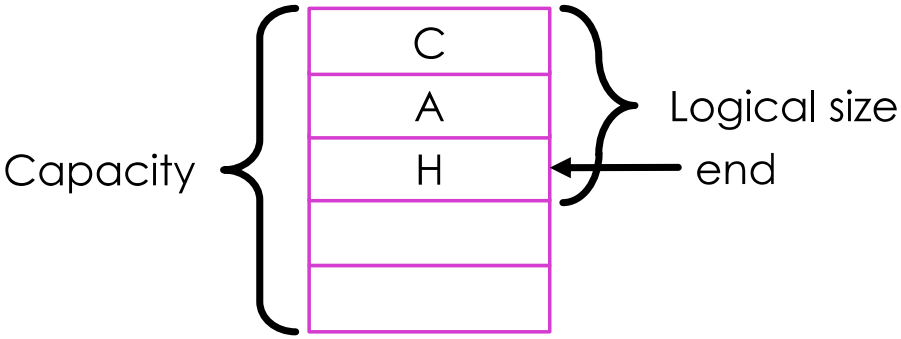
Arrays

- Fixed size: 0..n -1
  - Elements are identified by an index
- Oldest and very basic data structure
  - Close to the computer addressing scheme
  - Hardware support
  - Frequently used to implement other data structures
- Complexity
  - Insertion: O(1) -> a counter is usually employed
  - Insertion in a sorted array: O(n)
  - Search: O(n) when not sorted



Dynamic arrays

- Array that grows as you need it.
- "normal" array + some basic logic
  - Counter to track the end of the array
  - Strategy to allocate additional space when needed
  - Growing might need to copy all the data from the old smaller array to the new bigger array!
    - Insertion complexity: O(n)
- The growing factor depends on the implementation
  - Java ArrayList: 1.5 (from Wikipedia)
  - Python PyListObject: ~1.125 (n + (n >> 3)) (from Wikipedia)

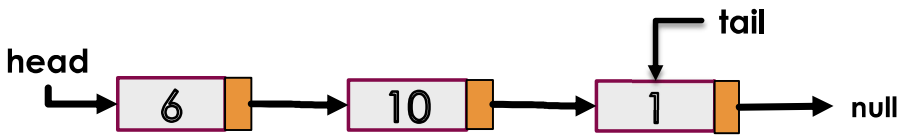


Linked Lists

- Collection of nodes holding minimum two fields
  - Data
  - Pointer to a **"next"** node
  - Data collection grows on demand
- The list starts with a **"head"** pointer
  - And usually ends in a **"tail"** pointer
- Insertion/removal
  - O(1) at the start
  - O(1) at the end if "tail" is present
  - In a sorted list: complexity is driven by the search algorithm

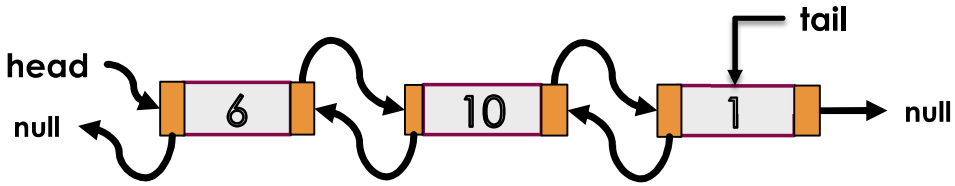
Linked Lists (cont)

Singly Linked List





Doubly Linked List





Circular Linked List

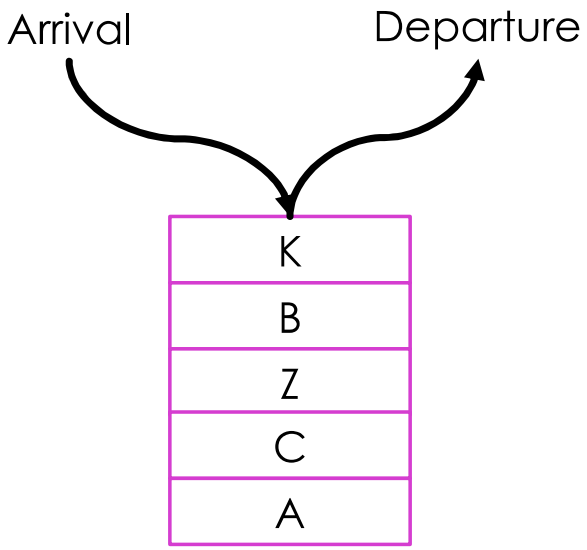
Hash Table

- Hash function is used to estimate the index where the data must be stored
  - Data to be stored is the function input
  - Insertion order is not preserved
  - Complexity: O(1) – in absence of collision, O(n) otherwise
- Collisions might arrive
  - Two different data might result in a same key
- Search complexity: O(1) in absence of collision
- Fixed size
  - Hash table growing implies to recompute, every key for every data already stored (update the hashing function)

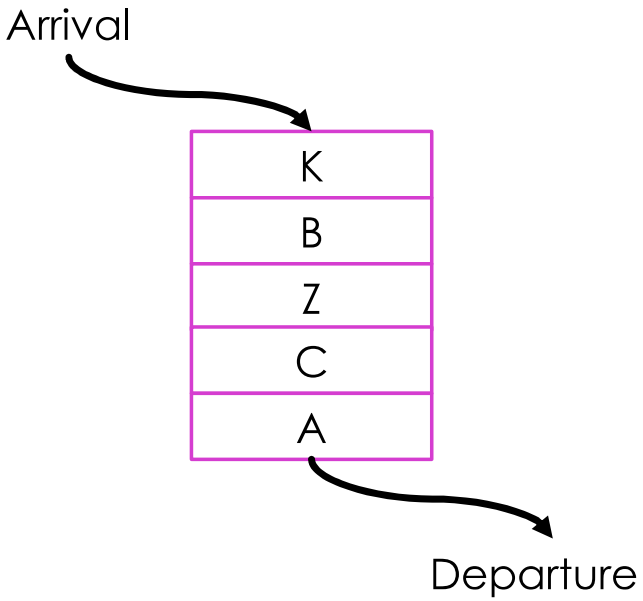
01	
02	
03	10
04	
05	
06	1
07	
08	5 -> 33 ->44
09	
0A	

Stack and Queues

Stack - LIFO



Queue - FIFO



Associative arrays

- Data structure to store paired keys and values
  - Real associative arrays are only possible with hardware support
    - Content Addressable Memory (CAM)
  - Frequently simulated by software (Python dictionaries)
    - Hash
    - Search trees
- Key might be any data type
  - The value might be a data type different to the key type

"bleu"	34523
"orange"	87645
"caramel"	56783
"melon"	34512

Trees

A lot to say about trees

But later…